



CLASS TEST PYTHON(18 FEB 2026)

Prepared By:

Name: Mahin Raj

Course: BCA (2nd Semester)

SAP ID: 590025881

Academic Year: 2025-2026

Submitted To:

Mr. Vibhu Gautam

Faculty, School of Computer Science

UPES, Dehradun



mahin-mr12 / python-programming



<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

python-programming / Test 18.ipynb



mahin-mr12 test

b4b9d3b · last month



175 lines (175 loc) · 3.26 KB

Preview

Code

Blame

Raw



```
In [40]: ##Q1
items = [("A", 90), ("B", 95), ("C", 95), ("D", 85)]
k = 2
def top_k(items, k):
    items.sort(key=lambda x: x[1], reverse=True)
    return items[:k]
print(top_k(items, k))
```

```
[('B', 95), ('C', 95)]
```

```
In [41]: ##Q2
def count_by_category(pairs):
    d = {}
    for cat, word in pairs:
        d.setdefault(cat, {})
        d[cat].setdefault(word, 0)
        d[cat][word] += 1
    return d
print(count_by_category(pairs))
```

```
{'sports': {'win': 1}, 'tech': {'ai': 1}}
```

```
In [33]: ##Q3
def missing_number(nums):
    n = len(nums)
    total = n * (n + 1) / 2
    return int(total - sum(nums))
```

```
In [34]: nums = [3, 0, 1]
print(missing_number(nums))
```

```
2
```

```
In [51]: def add_unique(x, seen=set()):
    if x not in seen:
        seen.add(x)
    return list(seen)
print(add_unique(1))
print(add_unique(2))
```

```
[1]  
[1, 2]
```

In [52]:

```
#10  
def parse(lines):  
    parsed_data = []  
    for lime in lines:  
        [  
            "John,25,90",  
            "John,25,80",  
            "Anna,130,70",  
            "Bob,20,105"  
        ]  
        return parsed_data  
{  
    "Jhon:25,85.0"  
}
```

Out[52]: {'Jhon:25,85.0'}

In []: